



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 01.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No: PHY F413

Course Title : Particle Physics

Instructor-in-Charge: Rishikesh Vaidya

Instructor(s): NA

Tutorial/Practical Instructors: NA

1. **Course Description:** (Just as every book has a preface, I will avail this space to elaborate about the course in great detail. It will be a synopsis of particle physics, providing context for the questions it addresses and a sense of the answers that will be discussed in the course. Hopefully, it will give you a fair idea of what particle physics is all about and what to expect from this course. The titles of paragraphs may seem out of place or even nonsensical in the first reading. Some things will fall in place as you read through the handout. For some other stuff, you may have to wait till the end of the course.)

Bright blue sky:

What are the most fundamental questions we can pose in Physics?

1. What are the most fundamental building blocks (called particles for the want of a better word) of the Universe?
2. What are the fundamental forces (interaction is a better word) that bind them to form the structural hierarchy we see at different levels?
3. What can we know about the particles and their dynamics if we know the answers to the above questions?
4. Are there fundamental principles that mandate the existence of different kinds of fundamental particles and different kinds of fundamental interactions? Is there an overarching theme in the answers we know?

Particle Physics aims to offer concrete answers to the above questions, though some caveats remain. Our present understanding of these questions is summarised in the so-called Standard Model (SM) of Particle Physics. This course will offer some answers within the framework of the Standard Model of Particle Physics.

Relativity + Quantum Mechanics = Quantum Field Theory:

Before I give a glimpse of answers to the above questions, we must ask for the framework for particle physics. It turns out that the natural framework for particle physics is a quantum theory for a system of fields. Let us understand the reason behind this. We will be talking about the most fundamental particles and their interactions at the minutest length scales, so quantum mechanics is inescapable. To study smaller and smaller length scales, we need probes whose wavelengths are sufficiently small. Thus, we use extremely high-energy particles (relativistic velocities imply larger momentum and hence smaller wavelengths). Hence, special relativity is a must. How about the study of mesons (bound states of quarks-antiquarks) and



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baryons (bound states of three quarks)? Here again, surprisingly, it is quantum mechanics that ensures relativistic conditions. Whenever you confine a quantum system to a tiny volume, uncertainties in the momenta become extremely large, implying relativistic conditions. Thus, quarks inside protons and neutrons are quite hot and hence relativistic. Now, under relativistic conditions, you can literally "cook" an arbitrarily large number of particles-antiparticle pairs, if you pump in a sufficient amount of energy. Thus, your underlying dynamical system must be inherently capable of describing situations involving an arbitrarily large number of particles. Thus, any framework with a fixed number of particles is doomed to fail. A system of fields with infinitely many degrees of freedom is the right dynamical system to describe relativistic interactions. Thus, relativity + quantum mechanics = Quantum Field Theory. Quantum Field Theory is demanding, and a typical graduate school has at least two courses in Quantum Field Theory before one can survey any literature in Particle Physics. However, the purpose of any undergraduate course in particle physics is essentially twofold: a) to provide a glimpse of the power of symmetry principles in the origin and classification of fundamental particles and their interactions, and b) to demonstrate the power of perturbative calculations of scattering cross-sections and decay rates via Feynman rules. To meet this end, we will follow an approach pioneered by Richard Feynman. This is the structure and approach of a first course in particle physics worldwide. It is just not possible to do squeeze in relativistic quantum field theory and particle physics in one semester, without any prior exposure to either of them.

From relativistic quantum mechanics to scattering cross-sections and decay rate:

Answers to questions 1 and 2 have been the content of the latter half of your Nuclear and Particle Physics course. The focus of this course would be essentially on questions 3 and 4. It turns out that we know a lot about the fundamental particles and their interactions. In the first module, we will primarily focus on calculating scattering cross-sections and decay rates. To do this, we will quickly recap the special theory of relativity (**module 1**) and the **time-dependent perturbation theory (module 2)**. In Module 2, we will also discuss the relativistic quantum mechanics of spin-0 particles (the Klein-Gordon equation). **Module-3** will introduce spin-1/2 particles (Dirac equation) and apply them to classic scattering cross-section calculations in quantum electrodynamics at lowest order in perturbation theory. You will see the famous Feynman diagrams and Feynman rules in action.. We will have some fun with what one can do with the rules, and that is calculating scattering cross-sections and decay rates, which experimentalists can actually measure.

Indiscreet fall of the discrete symmetries:

Symmetry transformations can be continuous or discrete. Discrete symmetries are essentially invariances under a discrete transformation and will be the subject of **module-4**. Their examples are parity (P) (inversion about origin, i.e., changing x,y,z to $-x,-y,-z$), time reversal (T) (changing t to $-t$), and charge conjugation (C) (changing particles to antiparticles and vice versa). Naively, Physics was expected to be invariant under each of these discrete transformations separately. However, it turns out that individually, each of them is violated. In fact, the so-called CPT theorem states that any Lorentz-invariant local quantum field theory with a Hermitian Hamiltonian must be invariant under the combined transformation of Charge conjugation (C), Parity (P), and Time reversal (T). It turns out that the weak interactions violate all three (C, P, and T) and also the combination CP. However, the combined CPT transformation is so far known to be an exact symmetry and any violation of CPT would signal a fundamental trouble either in quantum mechanics or relativity or local interactions. The Story of the separate violations of C, P, and T is intricately tied to the story of weak interaction and will be covered in **module 5**.

Building the Universe from scratch - the blocks and the glue:



Module-6 would be the last module of the course and will answer question number 4 in the affirmative and offer a convincing discourse on the same. Until the advent of Particle Physics in the 20th century, symmetries and conservation laws largely served a utilitarian purpose of bookkeeping. Conservation laws offer our first shot at dynamics without having to confront the dynamics head-on. Just look at the stuff before and after a dynamical event, and if conservation laws are applicable, we can pull out the unknown energy or momentum by a straightforward bookkeeping of conserved quantum numbers. However, during the latter half of the 20th century, we learnt that the symmetries aren't just utilitarian. They are the rhyme and the reason behind why things are the way they are.

What do you need to build the Universe from scratch? The bricks (fundamental particles), the cement (glue to bind them to form bound states), and a blueprint (the Lagrangian). All of these and much more are a consequence of some symmetry principles which can be broadly divided into two categories -- spacetime symmetries (Lorentz or, strictly speaking, Poincaré symmetries), and internal or local gauge symmetries. So the question of kinematics, or why the fundamental particles of different mass and spin exist at all, is answered by the spacetime symmetries. Particles of different mass and spin exist as representations of Poincaré group. On the other hand, the question of the existence of three different fundamental interactions (excluding gravity for the moment) is answered by the "internal" local gauge symmetries. They are "internal" in the sense that they operate in an abstract internal space different from the "external" spacetime. All the fundamental interactions are explained within the overarching theme of gauge theories.

How to break a symmetry without actually breaking it? A massive question:

If you think the question is an oxymoron and with a blatant misuse of adjectives, please bear with me until the mist clears. We now have the basic building blocks (particles) and the glue (interactions) which bind them in place as a consequence of spacetime (Poincaré) and internal (Gauge) symmetries. But what is the origin of the mass of the particles? It turns out that you cannot just put a mass term in your fundamental Lagrangian by hand and get on with the business. The question of mass is intricately tied to the question of Lorentz invariance and gauge invariance, and will be covered in the latter half of module 6. In a jigsaw puzzle, each piece is constrained by the shape of the neighbouring pieces. We cannot accommodate mass without fundamentally tweaking the particle content and introducing a "subtle breaking" of the gauge symmetries. There are three different ways in which a symmetry can be broken: explicit, spontaneous, and anomalous (beyond the scope of the course). It turns out that putting a mass term by hand explicitly breaks the local gauge symmetry, and that is nothing short of a sacrilege in particle physics. We can nevertheless break it through the "back door", that is, break it spontaneously! Our Lagrangian, equations of motion, commutation relations, and Feynman rules all respect gauge symmetry, and yet, the choice of a ground state violates the symmetry. Spontaneous symmetry breaking may appear mysterious, but it is rather common. The most famous example is that of a ferromagnet. Its Hamiltonian is invariant under rotational symmetry. This is why, above a critical temperature called the Curie temperature, thanks to thermal agitation, all the spin dipoles are randomly oriented and hence the net magnetization averages out to zero. However, below the Curie temperature, all dipoles spontaneously align in a specific direction, giving non-zero net magnetization. The Hamiltonian is still rotationally invariant, and yet magnetic dipoles select a specific direction and align. It turns out that to realize such spontaneous symmetry breaking in particle physics, we need to add a spin-0 field and its accompanying dynamics. How exactly the machinery of the Higgs mechanism and spontaneous symmetry breaking end up endowing fundamental particles with mass cannot be elaborated here. It is said that the proof of the pudding is in eating. We hope to get this far and discuss these issues.



Dark Clouds:

We started with bright blue skies and end with dark clouds. I wish I could end this preface on a positive note. However, truth be told, all of our knowledge about fundamental particles and their interactions is like a small lamp in a largely dark room. I am collecting here a laundry list of questions about which we have only speculations to offer without any hint of empirical verification.

1. **Dark matter and dark energy:** All of our understanding of the Universe encoded in the standard model of physics accounts for merely 5% of the observed Universe. The remaining 95 % consists of dark matter (27%) and dark energy (68%). The word here represents not only our ignorance about their origin and nature, but also that neither of these two forms of energy in the universe interacts directly with the known particles via known interactions. - except through gravity.
2. **Neutrinos are massive and mixed:** Neutrinos are massless within the standard model, but they are now known to have tiny non-zero masses, and the three species of neutrinos are also known to mix among themselves. To explain neutrino masses and mixing, we need to extend the standard model by introducing new particles and/or new interactions.
3. **Matter-antimatter asymmetry:** Although the laws of physics are largely symmetric under the particle-antiparticle exchange, the observed universe is grossly asymmetric. Theories that explain this asymmetry go by the name of baryogenesis (generating an excess of baryons over anti-baryons). There are scores of models, but none that have been empirically verified.
4. **The Flavour Problem:** We introduced the Higgs boson to explain the masses of fundamental particles, but the introduction of the entire Higgs sector is ad-hoc and poses more problems than it claims to solve. Why are there only three generations of quarks and leptons? Why is there a clear hierarchy of masses across generations? Why do masses take the value they take? The Standard Model cannot answer any of these questions.
5. **Gravity is excluded:** Notice that we have kept gravity out of the scheme of things. That is because gravity is very different from other forces in that we do not yet have a consistent and complete quantum theory of gravity that works at all energy scales.

There are a few more fundamental problems, but they are beyond the scope of this handout.

2. Scope and Objective of the Course: The above description conveys the scope and objective in great detail.

3. Text Books: Quarks and Leptons: An Introductory Course in Modern Particle Physics by Francis Halzen and Alan Martin, Published by Wiley India

4. Reference Books:

1. Introduction to Elementary Particles by David Griffiths, 2nd revised edition by Wiley-WCH publishing.
2. Modern Particle Physics by Mark Thomson, Cambridge University Press

5. Course Plan:

Module No.	Lecture Session	Reference	Learning outcomes
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1. (total lectures 5)	Relativistic Kinematics (L1.1-L1.5)	R1, 3.1-3.5, class notes	1. Define key terms such as four-vectors, invariant mass, and Lorentz factor and use invariant interval to find kinematic unknowns 2. Apply conservation of four momentum
2. (total lectures 7)	Time-dependent perturbation theory, rules for scattering amplitude (L2.1-L2.3), electrodynamics of spinless particles (L2.4-L2.7)	T1, 3.1-3.7, 4.1-4.7, class notes	1. Formulate the relativistic equation of a quantum state 2. Interpret negative energy solutions as belonging to an anti-particle state and appreciate the Feynman-Stueckelberg approach to negative energy solutions. 3. Formulate the problem of time dependent perturbation theory and translate the transition amplitude to various order in perturbation theory in the language of Feynman diagrams 4. Calculate decay rate and scattering crosssections for spinless particles
3. (total lectures 9)	Dirac equation (L3.1-3.5) and electrodynamics of spin $\frac{1}{2}$ particles (L3.6-3.9)	T1, 5.1-5.7, 6.1-6.7	1. Formulate relativistic equation for spin-1/2 particle and understand the degrees of freedom of a Dirac spinor and interpret probability density and current conservation. 2. Obtain free particle solutions, interpret negative energy solutions as anti-antiparticle, construct bilinear covariants, and examine zero-mass limit. 3. Calculate scattering cross-section for Moller and electron-muon scattering and Examine helicity conservation at high energies.
4. (Total lectures 4)	Discrete Symmetries (L4.1-L4.4)	R1, 4.4-4.4.4	1. Understand the basics of the group theoretical underpinnings of continuous and discrete symmetries 2. Explore violations of P(parity), C(charge conjugation), and T(time reversal) discrete symmetries in the weak interactions
5 (Total lectures 5)	Weak Interactions: Charged leptonic and quark weak interactions, neutral weak interactions (L5.1-L5.5)	R1, 9.1-9.2, 9.5-9.6	1. Calculate the decay rate of muon 2. Understand the structure of charged and neutral current weak interaction (V-A), 3. Appreciate the structure of chiral fermion state 4. Appreciate and understand electroweak unification and characterize the electroweak mixing via Weinberg angle



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6. (Total lectures 10)	Gauge Theories (L6.1-6.10)	R1, 10.1-10.9	1. Formulate for relativistic field theories 2. Understand gauge invariance for Abelian and non-Abelian fields 3. Understand spontaneous symmetry breaking and Higgs mechanism
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6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90 Min.	25%	<TEST_1>	CB
Comprehensive Examination	3 h	35%	<TEST_C>	OB+CB
Tutorials tests (Best 2 out of three)	50 min each	20%		CB
Project with Viva voce		20%		OB

7. Chamber Consultation Hour: Friday 5pm

8. Notices: Google classroom

9. Make-up Policy: Genuine cases of illness with convincing evidence and documents.

10. Note (if any): NC criteria less than 10% of Total marks.

Instructor-in-charge
Course No.